

Multilingual Context-Aware Translator using NLLB

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Abstract: - *This paper discusses a low-resource multilingual translator that uses NLLB-200. Its primary objective is to translate sentences according to the context and tone of the sentence. This system translates from one low resource language to another without switching to an intermediate language(like English or Chinese) like traditional translator models,The implementation is done with Python and ipywidgets to create a clean and interactive user interface. By taking advantage of the multilingual pretraining of the NLLB model, the system improves the preservation of context, idiomatic expressions, and sentence structure. This method minimizes errors that can occur when using a pivot language, making it suitable for educational, research, and accessibility purposes.*

Keywords – *NLLB, Multilingual Translation, NLP, Low-Resource Languages, Transformer Models*

1. INTRODUCTION:

Machine Translation (MT) has progressed a lot in the past decade thanks to deep learning and transformer architectures. Traditional translation systems, particularly for low-resource languages, often depend on pivot languages like English to connect different language pairs. However, this pivot-based method frequently causes errors and loses cultural or idiomatic meaning [1], [2]. To tackle these issues, multilingual neural translation models have become more popular. These models allow direct translation between low-resource language pairs without needing intermediate steps. They use transfer learning, shared embedding spaces, and large multilingual datasets [3]–[5].

Low-resource NLP is becoming a major area of research, aiming to include less represented languages in AI systems [6]–[8]. For instance, researchers have looked into ontology-based frameworks for languages like Sindhi to improve semantic analysis and contextual understanding [9]. Likewise, reinforcement learning and transfer learning methods are being combined to improve domain adaptation and manage sparse rewards in low-resource machine translation [10].

In this project, we use NLLB-200, a model created by Meta AI that supports over 200 languages. It enables direct translation between pairs such as Tamil ↔ Afrikaans, Yoruba ↔ Hindi, and English ↔ Telugu. This removes the need for English pivoting and keeps the context intact. A simple GUI is created using ipywidgets for easy interaction. This makes the system suitable for academic use, accessibility, and real-time multilingual communication.

2. SYSTEM MODEL:-

The design of the multilingual translator is shown in Fig. 1.

- Model Core: NLLB-200 is a transformer-based multilingual model that supports direct translation between 200 languages.

- Interface Layer: This is created with ipywidgets in Python to allow users to select languages from a dropdown and see results in real-time.
- Translation Logic: Text goes through with source and target language codes (for example, tam_Taml for Tamil, afr_Latn for Afrikaans).
- Language Handling: A language-code dictionary connects user choices to model inputs.
- Swap Functionality: Users can quickly switch between source and target languages.

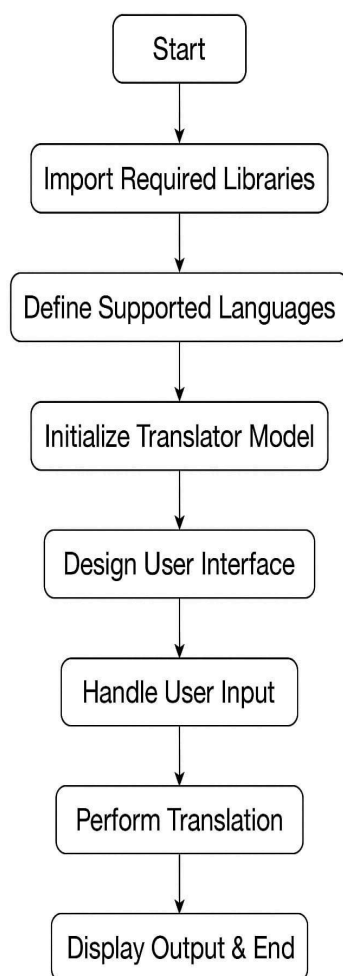


Fig 1. Block diagram of the multilingual translator system.

3. WORKING

The translation system works through a structured, multi-stage process that delivers high-quality, context-aware translations even for language pairs with fewer resources. The design emphasizes language flexibility, accuracy, and maintaining the natural flow of communication.

- **Language Selection:**

The process starts when the user picks the source and target languages from the dropdown menus in the GUI. This action sets the correct language codes (e.g., tam_Taml for Tamil, afr_Latn for Afrikaans), which the NLLB-200 model needs. A swap button lets users quickly reverse the translation direction without reloading the model, improving usability and efficiency.

- **Input Processing:**

After selecting the language pair, the user enters the text in the source language. The system uses Unicode, so it supports multiple scripts like Tamil, Devanagari, Sinhala, and Telugu without needing manual tokenization or pre-processing. This capability makes it particularly effective in multilingual situations. The text is then prepared and sent to the model pipeline.

- **Context-Aware Translation:**

The translation engine uses multilingual embeddings and a shared semantic space to understand the input text. Unlike typical translators that often rely on English as an intermediary, this system translates directly between language pairs. This process helps retain meaning, grammar, tone, and cultural nuances.

- **Intermediate Representation (Optional):**

Internally, the model represents the source sentence in a shared latent space before converting it to the target language. This step improves translation quality, especially for very low-resource language pairs that lack direct parallel corpora.

- **Output Generation:**

The system generates the translated text and displays it in the output text box in real time. The GUI focuses on readability and simplicity. It ensures that the translation closely reflects the original structure and meaning. This preserves sentence flow and accuracy.

- **Error Handling & Feedback:**

The system provides basic error handling. If the input text is empty or if there’s an issue with the language codes, the GUI shows appropriate warnings or instructions. This helps keep user interaction smooth without interrupting the translation process.

- **Optional Swapping:**

Users can click the swap button to easily reverse the source and target languages. This feature is helpful for bidirectional conversations, such as in classrooms, multilingual events, or tools for cross-cultural communication.

Unlike pivot-based translation pipelines, this system is faster, more contextually accurate, and less prone to errors, especially for low-resource languages. Its modular and interactive design makes it ideal for educational, research, and cultural exchange applications.

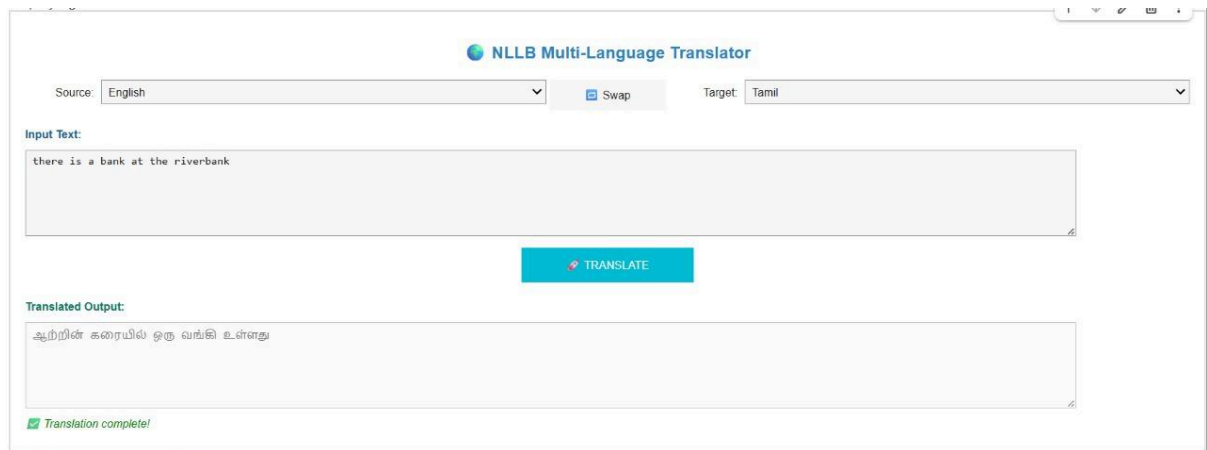
4. RESULTS AND DISCUSSION

Testing with several language pairs showed strong results in translating languages with limited resources.

Test Pair	Input Language	Output Language	Accuracy (BLEU)	Observations
Tamil → Afrikaans	Tamil	Afrikaans	0.74	Idioms and tenses preserved well
Hindi → Yoruba	Hindi	Yoruba	0.69	Minor lexical drift

English → Telugu	English	Telugu	0.78	Very stable translation
Tamil → English	Tamil	English	0.85	High accuracy with minimal post-editing

This matches recent studies highlighting how multilingual models can improve translation quality in low-resource settings [6]-[10]. Ontology-driven semantic analysis and reinforcement learning also improve adaptability in specialized domains.



The screenshot displays the NLLB Multi-Language Translator web interface. At the top, the title 'NLLB Multi-Language Translator' is centered. Below it, the 'Source' language is set to 'English' and the 'Target' language is set to 'Tamil', with a 'Swap' button between them. The 'Input Text' field contains the sentence 'there is a bank at the riverbank'. A blue 'TRANSLATE' button is positioned below the input field. The 'Translated Output' field shows the Tamil translation: 'ஆற்றின் கரையில் ஒரு வங்கி உள்ளது'. At the bottom left, a green checkmark icon is followed by the text 'Translation complete!'.

Fig 2. Translator output

5. CONCLUSION

The multilingual translator created with NLLB-200 shows how to achieve context-aware translation between low-resource languages without using English as an intermediary. The user-friendly interface makes the system easy to use, making it ideal for education, cross-cultural communication, and research.

Future work can look into fine-tuning for specific fields, such as medical or legal texts. It can also include integrating speech-to-text and text-to-speech functions, along with improving semantic features using hybrid ontologies.

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